

Igor Janotti

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SUMMARY

Software engineer with experience in distributed systems, networked services, and event-driven backend development. Strong foundation in data structures, algorithms, Linux fundamentals, and performance-focused debugging. Comfortable building reliable Python services, contributing to C/C++ systems, and working across APIs and cloud infrastructure. Interested in production AI/ML inference systems, latency optimization, and scalable model-serving platforms.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Java

Backend & Systems: Distributed systems, concurrent programming, event-driven architecture, networked services, API development

Infra & Platform: Linux, Docker, AWS (Lambda, DynamoDB, S3, SNS), Git, CI fundamentals, basic Kubernetes exposure

Observability & Reliability: Service debugging, system diagnostics, performance tuning, technical documentation

ML/Inference Exposure: LLM tool-calling workflows, prompt optimization, model-agnostic interface design

EXPERIENCE

Pacific Northwest National Laboratory

Jun 2025 – May 2026

Software Engineer Intern

Bellevue, WA

- Contributed to a distributed agent-based platform supporting real-world energy and building infrastructure
- Implemented networked services and backend components in Linux environments with a focus on reliability and fault tolerance
- Developed BACnet device discovery and scanning workflows to improve visibility across large sets of networked devices
- Built event-driven service features and optimized runtime behavior by debugging cross-service bottlenecks
- Collaborated with engineers to diagnose production-like issues, improve operational stability, and document service behavior

Bellevue College

Feb 2024 – Mar 2024

Teaching Assistant – Programming Languages (C++)

Bellevue, WA

- Assisted students with debugging and systems-level reasoning, including memory behavior and execution flow

PROJECTS

VOLTTRON AI: Conversational Smart-Grid Control Interface | *Python, Pydantic AI, LLMs*

- Built a model-agnostic conversational interface that maps natural language requests to structured platform operations
- Implemented modular tool-calling and validation layers for configuration, diagnostics, and platform control actions
- Improved responsiveness and reliability with cached path resolution, singleton environment management, and streaming outputs

Cloud-Based Event-Driven Processing Pipeline | *C#, AWS*

- Designed an event-driven microservice-style pipeline for ingestion, processing, and storage across cloud services
- Built asynchronous workflows to support higher throughput, dependable processing, and clear service boundaries

FollowBot Companion Robot | *C++, Python*

- Implemented real-time control and sensor feedback loops, balancing responsiveness with stable system behavior

EDUCATION

Bellevue College

Bellevue, WA

Bachelor of Science in Computer Science

August 2025

- GPA: 3.54